

Assessment of Ionospheric Spatial Decorrelation for Daytime Operations of GBAS in the Brazilian Region

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ABSTRACT

Extensive ionospheric studies were conducted to support the initial phase of system design approval (SDA) for the existing SLS-4000 GBAS installed at Galeão Airport (GIG) in Rio de Janeiro, Brazil. This paper focuses on determining the broadcast value of the standard deviation of vertical ionospheric gradients (or σ_{vig}) that is required to bound ionospheric spatial gradients at the Brazilian region under nominal conditions during daytime hours. The days for the analysis were selected by a combination of high levels of daytime or nighttime scintillation and/or severe values of the geomagnetic storm index. The time-step method, which is useful for gaining sufficient samples at distances less than the physical separation distance of ground stations, was utilized to estimate ionospheric spatial gradients. The results show that the bounding σ_{vig} values are in the range of 8-13 mm/km, when daytime was defined to exist between 6 AM and 6 PM local time. Because the time-step method estimates gradients over a time interval instead of at a single time, the results include temporal as well as spatial gradients, meaning that they overestimate the spatial gradients that are the desired results. Thus, a new method called “geometric modeling” was proposed to estimate ionospheric temporal gradients and evaluate the temporal effect added to the bounding σ_{vig} values. As a result, a σ_{vig} of 13 mm/km including an approximately 2 mm/km temporal gradient contribution, is conservative enough to bound ionospheric spatial decorrelation for daytime GBAS operations in the Brazilian region.

1.0 INTRODUCTION

The Honeywell SLS-4000 GBAS, which was originally configured with the ionospheric parameter and threat model for the conterminous United States (CONUS), was installed at Galeão Airport (GIG) in Rio de Janeiro, Brazil. Ionospheric activity in equatorial regions (low-latitude regions within 25 degrees latitude of the geomagnetic equator) including the Brazilian region is known to be significantly more variable and intense than that encountered (and extensively studied) in mid-latitude regions such as CONUS. Thus, the CONUS ionospheric parameter and threat model needed to be validated before applying them to GBAS in Brazil. Previous work [1] found that many anomalous gradients significantly exceeded the CONUS threat model bounds in the Brazilian region (specifically, 31 validated gradients were observed that exceeded 500 mm/km, with a maximum of 850.7 mm/km). However, all of these gradients occurred during local nighttime (with only a handful of exceptions), when disturbed ionosphere and plasma bubbles may occur. The few events that occurred outside local nighttime did not exceed the CONUS threat model bounds (375 – 425 mm/km). The results of [1] suggested that, while additional ionospheric mitigation would be needed for CAT I GBAS to be provided at nighttime, the existing SLS-4000 with the CONUS threat model might be sufficient to support operations during daytime. As an extension of the ionospheric threat model study [1] based upon Brazilian ionospheric data from 2011 to 2014, additional days of recent severe ionospheric storms during March 2015 and September 2017 and well-known storm days during the previous Solar Cycle 23 in 2003 and 2004 were examined [2]. This study also confirmed that the CONUS threat model covers all gradients that have been observed during daytime hours.

While no additional ionospheric threat mitigation is needed beyond the existing SLS-4000 during daytime hours, the nominal standard deviation of vertical ionospheric gradient (σ_{vig}) over Brazil should be assessed to reflect low-latitude ionospheric conditions, which are different from those in mid-latitude regions. Small user errors remain after GBAS corrections are applied due to ionospheric spatial decorrelation between reference station and users. This spatial decorrelation error should be taken into account when the user computes protection levels (PL) which bound the true position error to the required integrity probability [3]. This is

done in the calculation of PL by using the broadcast σ_{vig} , which is selected to meet the integrity requirements of CAT I precision approaches. In mid-latitude CAT I GBAS operations, such as in CONUS, the value of σ_{vig} that bounds nominal ionospheric spatial gradients has been determined to be 4.0 mm/km [3]. The actual value that is broadcast by the SLS-4000 is the Root-Sum-Square (RSS) of this value and a separate value of 5.0 mm/km that was conservatively chosen to bound anomalous tropospheric gradients. Therefore, the minimum broadcast value of σ_{vig} in CONUS is $(4.0^2 + 5.0^2)^{0.5} \cong 6.4$ mm/km. The same approach can be taken in Brazil, but the larger spatial gradients created by equatorial ionospheric behavior will require a higher value of the ionospheric component of σ_{vig} .

This paper determines the minimum value of σ_{vig} which is required to bound ionospheric spatial gradients at GIG under nominal conditions during daytime hours. Section 2.0 describes the datasets used to estimate nominal ionospheric spatial gradients in the Brazilian region. Section 3.0 explains the time-step method that was used to estimate ionospheric spatial decorrelation. Section 4.0 gives estimates of the bounding σ_{vig} over Brazil during early morning, daytime, and early evening hours and recommends a specific value of σ_{vig} that is valid over a recommended local daytime window. Section 5.0 explains how to translate this general guidance regarding the duration of the daytime window into specific times to begin and end SLS-4000 operations that can be derived (as a function of season) from GPS Time. In Section 6.0, the temporal gradient effect added to the bounding σ_{vig} is evaluated by estimating temporal gradients via a newly proposed “geometric modeling” method. Finally, Section 7.0 draws conclusions with some recommendations for future work.

2.0 DATA

2.1. Data from Brazilian GNSS Networks

The work utilizes data collected for the Phase I Brazilian threat model study that was completed in 2015 [1]. This study analyzed data from several networks of ground stations in South America, including the Brazilian Network for Continuous GPS Monitoring (RBMC), the Low-latitude Ionospheric Sensor Network (LISN), the Concept for Ionospheric scintillation mitiGAtion for professional GNSS in Latin America (CIGALA), the ICEA Septentrio and Trimble receiver network, the Integrated Positioning System for GeodynamicStudies (SIPEG), and the Honeywell SLS-4000 receivers installed at and near Galeão airport (GIG). Figure 1 shows the locations of the approximately 185 GNSS reference stations used in this study (as of 2014), color and symbol-coded by the network that each station belongs to. The densest region of reference stations corresponds to the most populated regions of Southeastern and Coastal Brazil. A total of 123 days of data during a three-year period from April 2011 to March 2014 were selected for detailed study based on a combination of days with strong nighttime scintillation (85 days), space weather intensity as measured by global K_p values (30 days), and a small set of quiet days for comparison (8 days).

2.2. Date Selection

Ionospheric scintillation is one of low-latitude features distinguished from the mid-latitude ionospheric behavior. Scintillation events are known to occur during nighttime mostly, but they can extend toward the early morning and late-afternoon hours. Huang et al. [4] reported that plasma bubbles, which generally occur at nighttime in the equatorial ionosphere and often coincide with scintillation, could have a long lifetime of 12 hours and exist during daytime until mid-afternoon. Thus, the days for the analysis were selected by a combination of high levels of daytime or nighttime scintillation (as measured by the “Scintillation Occurrence Rate” or “SOR”) and/or severe values of the Dst index. SOR is the ratio of the number of points in a time period in which the S_4 amplitude scintillation index exceeds a threshold to the total number of points in that time interval, and is defined as follows:

$$\text{SOR} = \{ N[S_4 \geq 0.2] / N[\text{all}] \} \times 100\%$$

where $N[\text{all}]$ is the total number of points observed within a certain time period and $N[S_4 \geq 0.2]$ is the number of points in that time interval in which the S_4 amplitude scintillation index exceeds 0.2, indicating a significant degree of amplitude scintillation. The database of 123 ionospheric active days during a three-year period from April 2011 to March 2014 from [1] was utilized for this data selection. The two ionospheric data sets were selected based on the highest daytime SOR and the highest nighttime SOR. The lists of the selected days, their data categories from Phase I study, SOR with rank, and the overbounding σ_{vig} results are shown in Tables 1 (the top five daytime SORs) and 2 (the top six nighttime SORs).

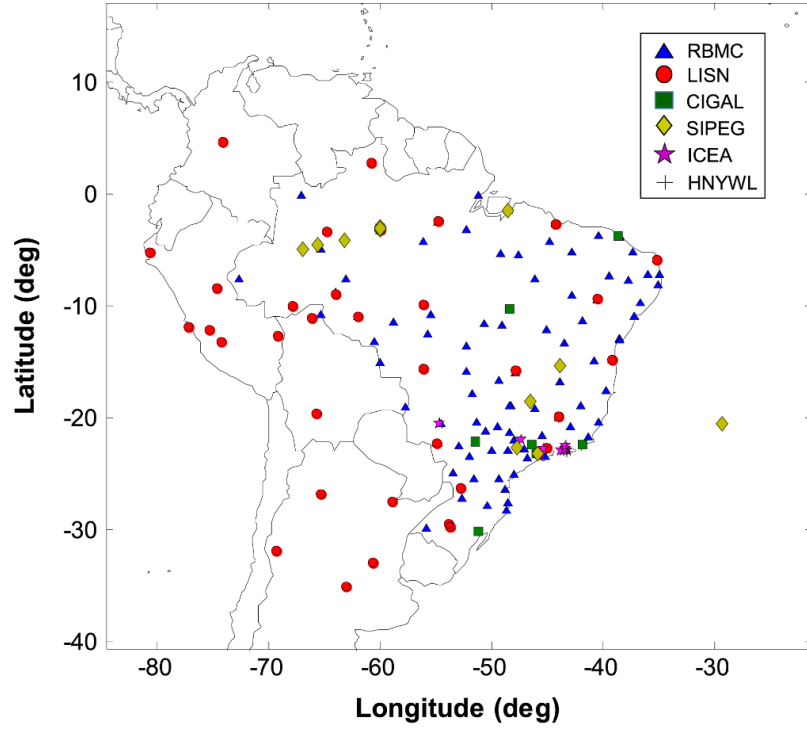


Figure 1: GNSS reference stations and networks used in Phase I ionospheric threat study [1]

Table 1. Dataset 1 days and results: Selected based on highest daytime SOR

ID	Date	Phase I Category	Daytime SOR (Rank) (%)	Overbounding σ_{vig}
1	March 7 th , 2012	Selected by Dst	2.1141 (1 st)	10 mm/km
2	January 25 th , 2012	Selected by Dst	2.1048 (2 nd)	9 mm/km
3	March 9 th , 2012	Selected by Dst	0.7714 (3 rd)	9 mm/km
4	October 25 th , 2011	Selected by Dst	0.6875 (4 th)	10 mm/km
5	October 8 th , 2012	Selected by Dst	0.5837 (5 th)	8 mm/km

Table 2. Dataset 2 days and results: Selected based on highest nighttime SOR

ID	Date	Phase I Category	Nighttime SOR (Rank) (%)	Overbounding σ_{vig}
6	October 1 st , 2012	Selected by Dst	18.3265 (1 st)	N/A (Anomaly day)
7	March 3 rd , 2014	Scintillating day	17.4038 (2 nd)	12 mm/km
8 (=4)	October 25 th , 2011	Selected by Dst	14.9240 (3 rd)	10 mm/km
9	March 4 th , 2014	Scintillating day	14.8806 (4 th)	12 mm/km
10	February 25 th , 2014	Scintillating day	14.4754 (5 th)	13 mm/km
11	February 26 th , 2014	Scintillating day	13.3409 (6 th)	11 mm/km

3.0 IONOSPHERIC GRADIENT ESTIMATION METHOD

The two methods, station-pair method and time-step method, were developed and used to estimate anomalous ionospheric spatial gradients in previous work [3, 5]. The station-pair method used to estimate anomalous ionospheric gradients is illustrated in Figure 2 (left). Estimates of ionospheric slant delay are made by two ground stations (in known locations) from the same GPS or GNSS satellite at the same time, and the spatial gradient is calculated simply as the difference between these two delays divided by the baseline distance (on the ground) between two stations. Note that distance within the presumed ionospheric thin shell are deliberately not used, as the thin-shell height is much more uncertain under anomalous conditions. For a similar reason, slant ionospheric gradients are used instead of using the ionospheric obliquity factor (derived from the thin-shell model at a specific height) to calculate elevation-independent vertical or “zenith” gradients.

The time-step method is also used to estimate ionospheric gradients, and it is shown in Figure 2 (right). It only requires one ground station, and it estimates gradients over the path in space created by satellite motion from ionospheric pierce point (IPP) at t_1 to IPP point at t_2 over a short time interval, $\Delta t = t_2 - t_1$. Here, the distance over which the change in ionospheric delay is divided by is based on the presumed IPP separation distance based on the ionosphere being concentrated in a thin shell at a particular altitude (typically 350 km). As noted above, this is a poor assumption under anomalous conditions, but it is better under nominal conditions. Also, because gradients are estimated over a time interval instead of at a single time, the results include temporal as well as spatial gradients, meaning that they overestimate the spatial gradients that are the desired results. Given these limitations, the time-step method is used only for estimating gradients on nominal days (as done in [3], accepting that the results will be conservative), validating anomalous gradients estimated using the station-pair method (as done in [6]), and as a backup method for estimating anomalous gradients when the density of ground stations is insufficient to provide good estimates from the station-pair method.

In this study, the time-step method was used to estimate ionospheric gradients, since the density of South American ground stations from which data was collected and analyzed is insufficient to provide good estimates from the station-pair method. This method is useful for gaining sufficient samples at distances less than the physical separation distance of ground stations, especially when the baseline among nearby stations are too far apart. The time window ($\Delta t = t_2 - t_1$) of one minute was used in this study to obtain the spatial distance of interest (i.e., short distances of about 5 km between the location of decision height and ground facility for the GBAS application). However, ionospheric temporal decorrelation over one minute interval would be included in the spatial gradient estimates, which will be further discussed in Section 6.

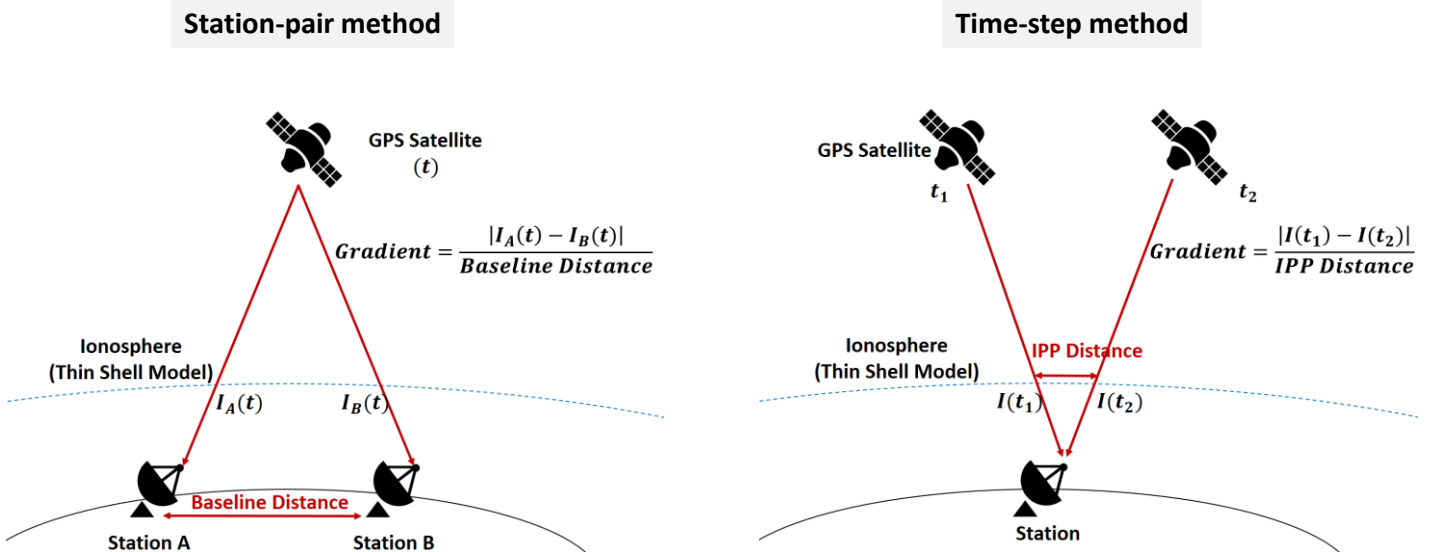


Figure 2: Station-pair method for estimating anomalous ionospheric gradients (left) and time-step method for validating anomalous ionospheric gradients and estimating nominal ionospheric gradients (right)

4.0 OVERBOUNDING NOMINAL σ_{vig} FOR BRAZILIAN REGION

4.1. Overbounding Method

In this study, the sigma overbounding method [3] was used to assess the nominal σ_{vig} value for Brazilian region. Figure 3 shows the spatial decorrelation result for the day (7 March, 2012) with the highest daytime SOR value, which is ID 1 from Dataset 1 in Table 1, using the time-step method and the Brazilian GNSS network data. The difference in vertical ionospheric delays is shown as a function of the IPP separation distance. The time window, Δt , of one-minute corresponds to the IPP separation distance of about 3 to 8 km. The absolute value of the vertical delay difference increases fairly linearly as the IPP separation distance increases. The differential delays were divided by the corresponding IPP separation distances to obtain vertical ionospheric gradients.

The one minus Cumulative Distribution Function (1-CDF) of normalized vertical ionospheric gradients is shown in Figure 4. The vertical gradients are normalized by removing their mean and dividing them by their standard deviations derived from the observations shown in Figure 3. Figure 4 shows that the distribution derived from the observations has non-Gaussian tails. Because GBAS users assume a zero-mean normally distributed error model in the computation of protection levels, the nominal sigma (1σ) of a zero-mean Gaussian distribution (shown as the dashed curve) should be inflated to cover the heavy tails of the actual distribution (the red curve). We inflate the nominal sigma until the 1-CDF of normalized vertical gradients hits that of the normal Gaussian distribution. The inflation factor needed for the 7 March, 2012 data was 1.134. The 1-CDF of gradients normalized with inflated sigma (the blue curve) falls well below that of the zero-mean normal distribution (the dashed curve), meaning that the inflated Gaussian distribution well overbounds the empirical distribution.

The sigma overbounding method is used as follows. The mean and standard deviation of vertical ionospheric gradients in each bin of IPP separation distance are computed, interpolated at each distance corresponding to each gradient, and used to normalize the gradients. Based on the distribution of normalized ionospheric gradients, the inflation factor (f) is then determined as shown in Figure 4. Lastly, the overbounding σ_{vig} is computed as $|\mu_{\text{vig}}| + f * \sigma_{\text{vig}}$ for each bin. Figure 5 shows the σ_{vig} overbound result on 7 March, 2012. The estimated σ_{vig} overbound (the pink curve) at shorter IPP separation distances less than 4 km cannot be trusted because temporal gradients degrade the results. The estimates at distances larger than 6 km were obtained using insufficient number of data for reliable statistics. The results obtained at IPP separation distances of 4 to 6 km supports that σ_{vig} is conservatively bounded by 10 mm/km on 7 March, 2012.

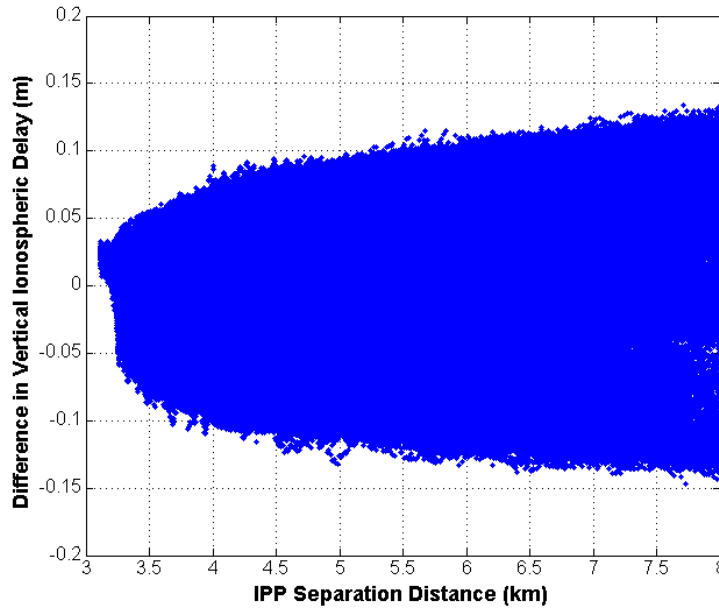


Figure 3: Differential vertical ionospheric delay results on the highest daytime SOR day (7 March, 2012) from the time-step method and Brazilian GNSS network data.

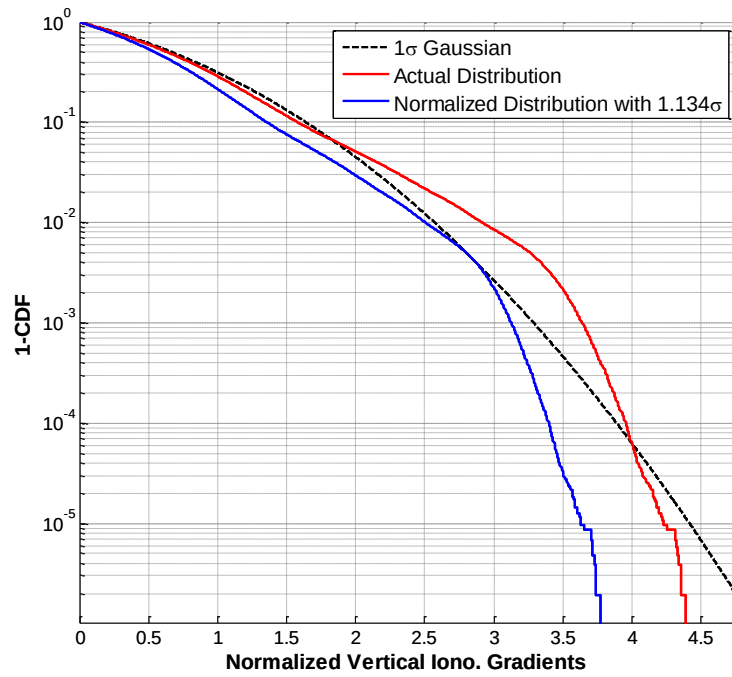


Figure 4: 1-CDF of normalized vertical ionospheric gradients on the highest daytime SOR day (7 March, 2012)

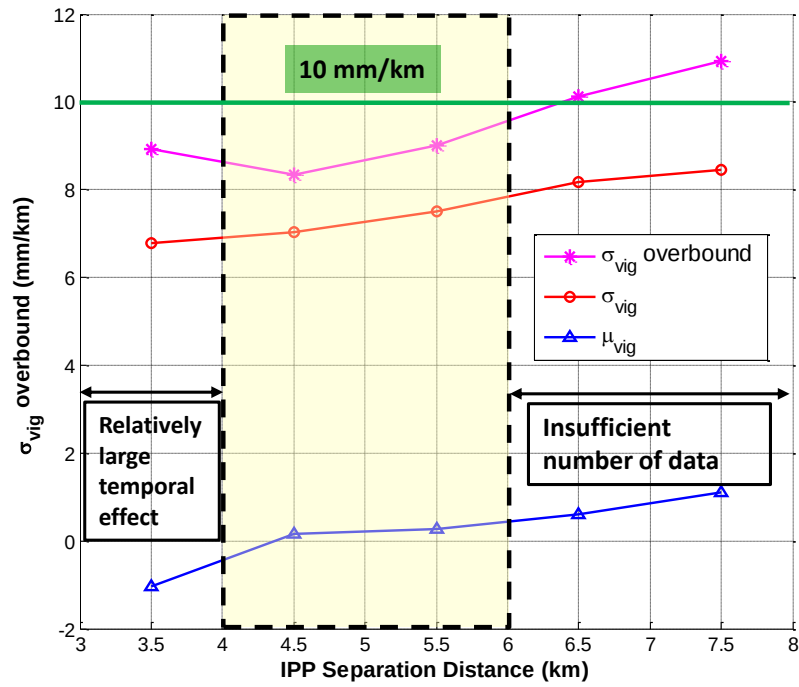


Figure 5: σ_{vig} overbound results from the time-step method and Brazilian GNSS network data on the highest daytime SOR day (7 March, 2012)

4.2. Overbounding σ_{vig} Results for Selected Datasets

The analysis in this study examined the two ionospheric data sets from the Phase I threat analysis database as explained in Section 2 using an approach similar to that of [3] but using the time-step method to derive spatial gradient estimates in place of the station-pair method, which requires a denser network of ground receivers. A summary of the days included in these two data sets and the overbounding σ_{vig} results for each day are shown in Table 1 and 2. One of the days identified, 1 October, 2012 in Dataset 2 (ID 6), turned out to be an anomalous day because it included gradients exceeding 100 mm/km as shown in Figure 6. The other days, analyzed independently, generally produced bounding σ_{vig} values in the range of 8 – 13 mm/km, when daytime was defined to exist between 6 AM and 6 PM local time, and where “ σ_{vig} ” here refers only to the nominal ionospheric component and not to the RSSed tropospheric component.

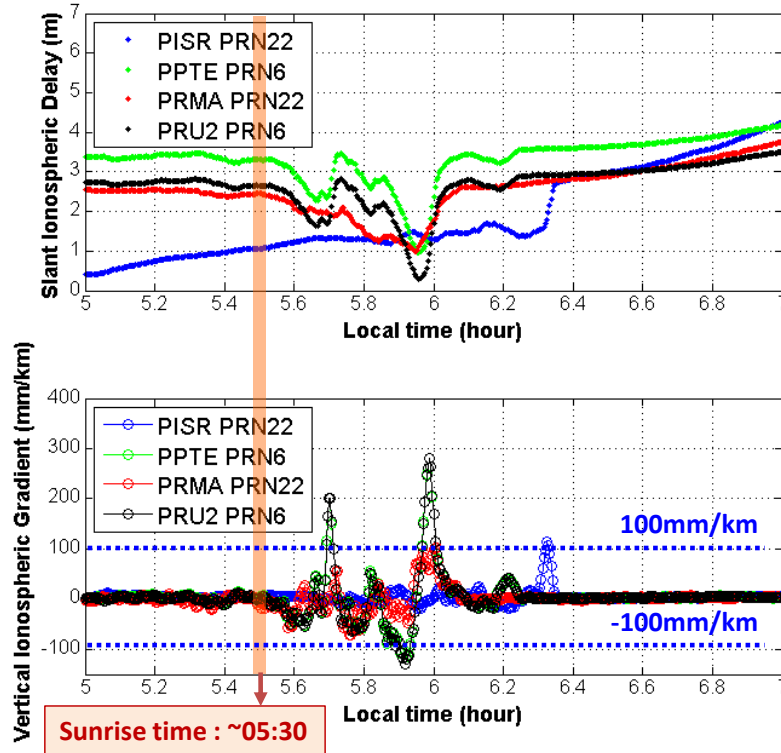


Figure 6: Slant ionospheric delay and vertical ionospheric gradient estimates for four satellite-station pairs on 1 October, 2012

As expected, the largest values of σ_{vig} occur at the edges of daytime, meaning the periods before and around local sunrise and around and after local sunset. Figure 7 and Figure 8 illustrate this on October 25th, 2011. Figure 7 compares vertical (zenith) differences in ionospheric delay as a function of IPP separation distance for data sets starting at 6 AM local time (LT) (in blue) and 5 AM LT (in red). Much larger variations appear in the data that starts at 5 AM LT. Figure 8 shows that this was caused by gradients exceeding 100 mm/km occurring prior to sunrise (at 5:50 AM LT) and affecting multiple satellites viewed from station POAL (Porto Alegre). More generally, the result on Figure 9, that combines 9 days of data and computes σ_{vig} separately for each two-hour time bin shows that the periods from 4 to 6 AM LT and 6 to 8 PM LT (just beyond daytime) have much higher σ_{vig} values than the periods between 6 to 8 AM LT and 4 to 6 PM LT.

For this reason, we recommend that the definition of daytime for GBAS operations should correspond to the period between 6 AM and 6 PM LT. Because this is a general guideline based upon data analyzed from throughout Brazil, and because the beginning and end of daytime for ionospheric purposes is most closely related to sunrise and sunset times (which vary throughout the year), more specific guidance is required for the SLS-4000 at GIG. This is addressed further in Section 5.0.

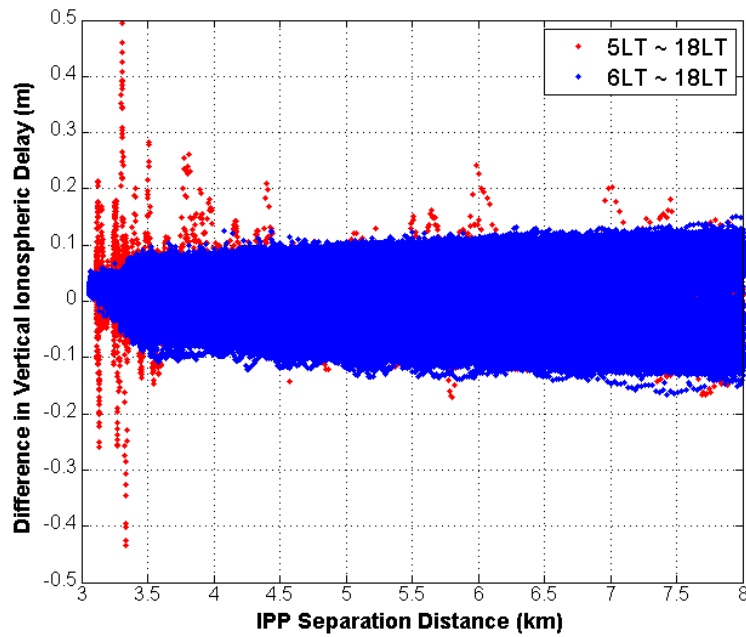


Figure 7: Differential vertical ionospheric delay vs. IPP separation distance on 25 October, 2011

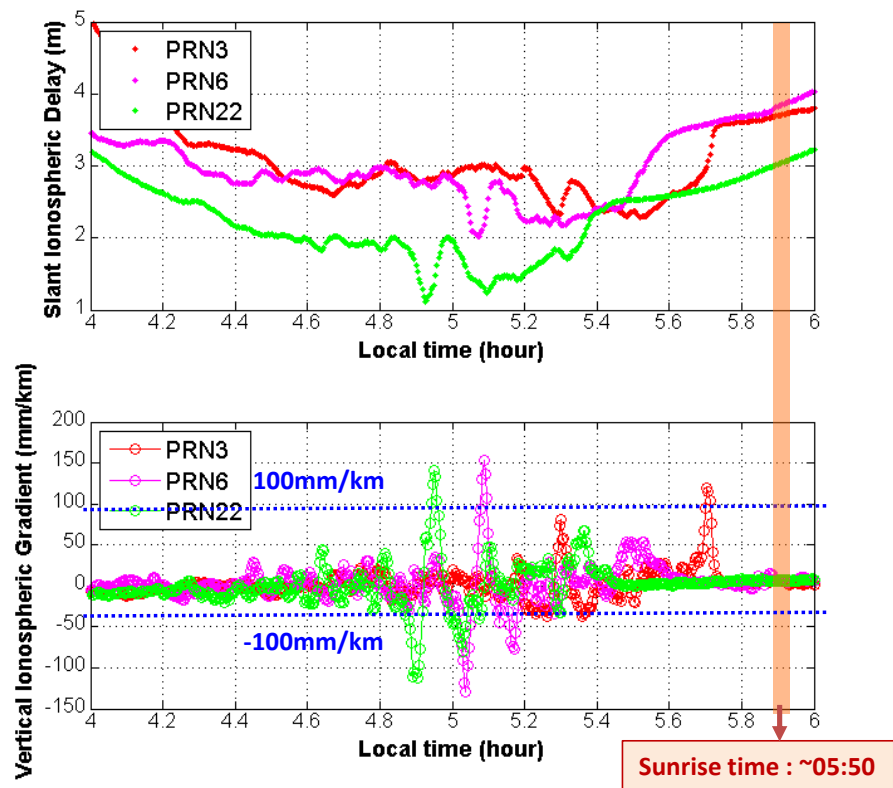


Figure 8: Ionospheric behavior vs. Local time for several satellites at POAL on 25 October, 2011

Using the entire Brazil database within 6 AM and 6 PM LT, Table 2 and Figure 10 all suggest that the bounding ionospheric contribution to the nominal broadcast σ_{vig} should be 13 mm/km. Table 2 for Dataset 2 identifies 25 February, 2014 as the day with the highest σ_{vig} when σ_{vig} is computed separately for each day, and that highest value is 13 mm/km. In Figure 10, which is a zoomed-in version of Figure 9, when σ_{vig} is computed over 9 days (including February 25th, 2014) but is broken down into two-hour intervals, the maximum σ_{vig} value within daytime is also 13 mm/km. Selecting 13 mm/km as the ionospheric component of the nominal σ_{vig} and performing an RSS with the same 5 mm/km tropospheric contribution as in CONUS results in a minimum broadcast value of 13.93 mm/km, which should be rounded up to 14 mm/km.

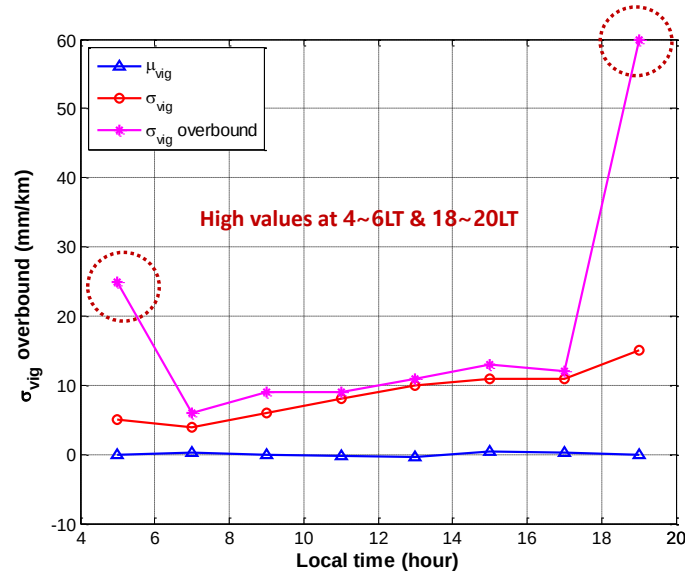


Figure 9: Overbounding σ_{vig} vs. Local time for nine days of combined data

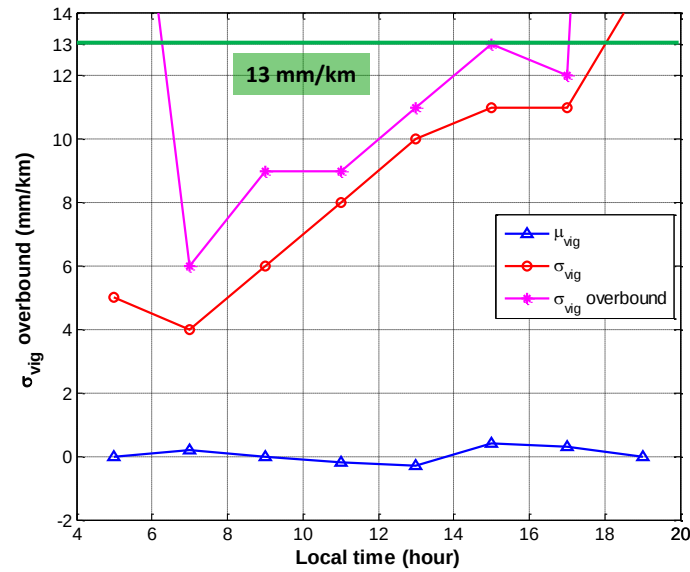


Figure 10: Zoomed-in view of overbounding σ_{vig} vs. Local time for nine days of combined data

5.0 DAYTIME DEFINITION

As explained above, the definition of “daytime” hours as between 6 AM and 6 PM LT is really an approximation of local sunrise and sunset times, as ionospheric behavior violating the σ_{vig} value proposed above appears to be driven by sunrise/sunset rather than specific local times. Therefore, the SLS-4000 installation at Rio should use an approximation of local sunrise and sunset times, which vary throughout the year, instead of specifically using 6 AM and 6 PM LT. Since the SLS-4000 naturally observes GPS Time instead of local time, the daytime transition times need to be referenced back into GPS Time to be implemented within the SLS-4000.

Various online almanacs exist for computing sunrise and sunset times for particular locations and times (see [7] and [8]). Figure 11 was produced by [7] and shows the times of sunrise and sunset (“Daylight”) in Rio de Janeiro for the year 2018 as a function of local time, including the shift to and from Summer Time in February and November. The trend of shortening and lengthening days by season is evident here. The detailed tables provided by [7] show that the maximum difference in sunrise times between solstices is about 1.47 hours (6:04 AM “standard” time on 21 December vs. 7:32 AM “standard” time on 21 June). A review of tables generated for the next 20 years shows that these times change very little, so a daytime conversion table built based on 2018 should be valid for the lifetime of an SLS-4000 at Rio.

This variation in sunrise and sunset times needs to be provided to the SLS-4000 in a convenient format along with a means for the SLS-4000 to convert between local time, UTC, and GPS Time. For now, it will be assumed that the latter is already present within the SLS-4000 (this will be revisited if a means for conversion needs to be provided). A table of local sunrise and sunset times can be created and input into the SLS-4000 at varying levels of precision (e.g., one entry per month), ignoring the shifts caused by Summer Time (as these do not affect GPS Time). An alternative is to fit polynomial curves to the actual sunrise and sunset times (as a function of day of year) to allow the SLS-4000 to compute estimates of these times for each day. Precise accuracy in computing sunrise and sunset times is not required: results within ± 5 minutes of the actual times should be sufficient. To the degree possible, these models should be developed offline to minimize the new calculations required of the SLS-4000.

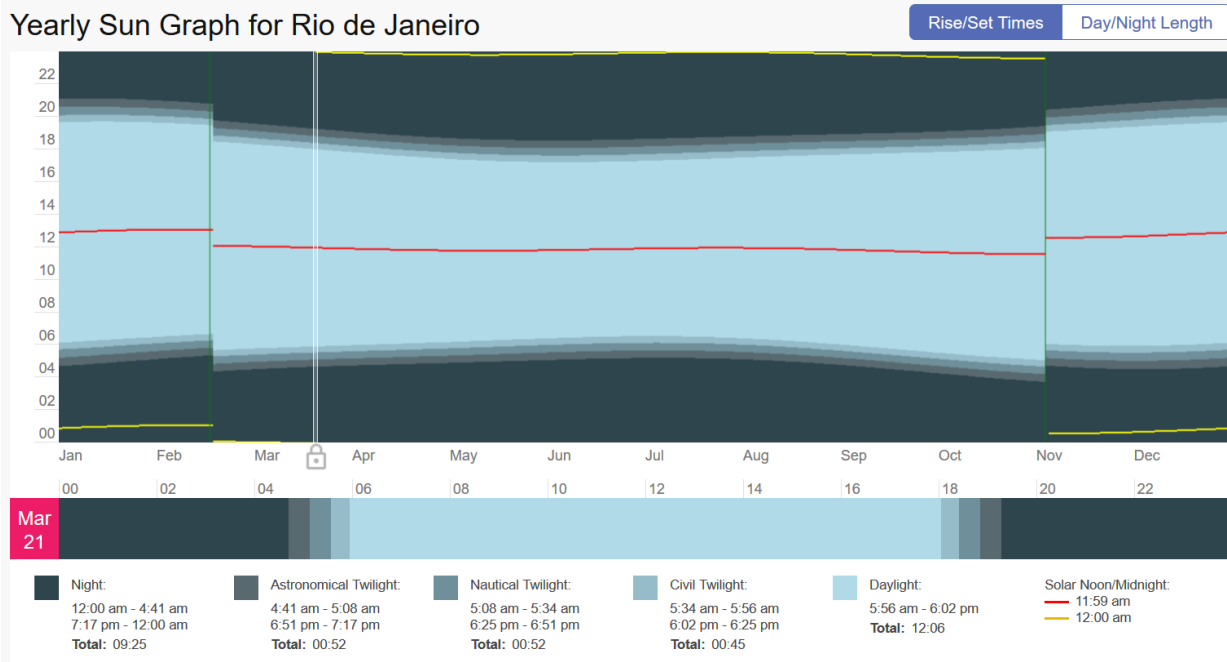


Figure 11: Sunrise and sunset times for Rio de Janeiro in 2018 [7]

One other consideration is that the days used to establish the minimum nominal σ_{vig} and the daytime period in this study were all between September and April, corresponding to the more active months of the year for the ionosphere in South America. These months also correspond to days with longer daytime periods, as shown in Figure 11. By using the results from these months during the quieter mid-year period (Brazilian winter) when the days are shorter, we may be overly limiting the achievable GBAS

performance, meaning that the minimum broadcast σ_{vig} could be lower and/or that the definition of “daytime” could be extended while still being protected by the minimum broadcast σ_{vig} and CONUS threat model. Additional data analysis of mid-year days would be needed to verify this hypothesis.

6.0 TEMPORAL GRADIENT EFFECTS

6.1. “Geometric modeling” for Temporal Gradient Estimation

This study evaluates the temporal gradient effect added to the bounding σ_{vig} introduced by the time-step method. A new method called “geometric modeling” is proposed to estimate temporal gradients. The geometric model describes differential ionospheric delay computed by the time-step method as a linear combination of spatial and temporal delay differences. If it is possible to extract the component of spatial delay difference from the total delay difference, ionospheric temporal decorrelation can be estimated. The underlying assumption to make this possible is that the ionospheric decorrelation between two points can be measured by both the station-pair method and the time-step method. Figure 12 shows the configuration of the geometric modeling. The “ t_1 Ionosphere” is the ionosphere state at time t_1 , and the “ t_2 Ionosphere” is the ionosphere state at time t_2 , which is after Δt from t_1 . S_1 denotes the ionospheric delay measurement from the pair of satellite at t_1 and receiver RV_1 , and S_2 is the ionospheric delay measurement from the pair of satellite at t_1 and receiver RV_2 . P_2 is the ionospheric delay measurement from the pair of satellite at t_2 and receiver RV_1 . We assume the IPPs of S_2 and P_2 are very closely located and thus can measure ionospheric delays at the same location.

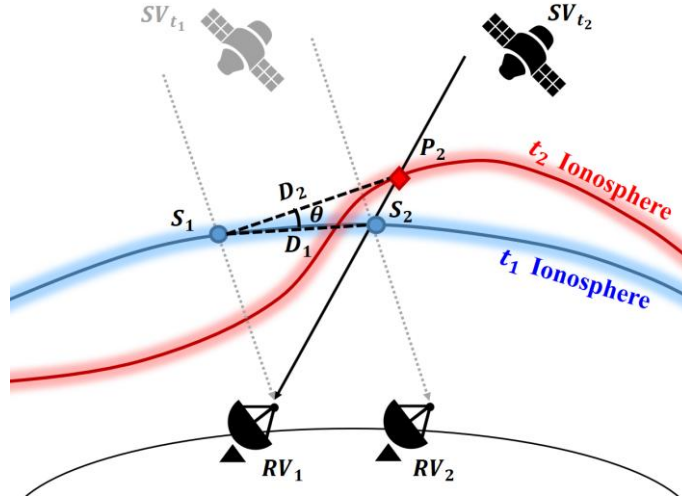


Figure 12: Configuration of the “geometric modeling”

The time-step method was used for estimating ionospheric spatial gradients in Section 4. If we use the symbols described above, the differential delay obtained using the time-step method can be written as

$$dI_{\text{time-step method}} = P_2 - S_1. \quad (1)$$

Assuming that P_2 and S_2 are delay estimates at the same location but at different times with a time interval of $\Delta t = t_2 - t_1$, the ionospheric delay difference caused by temporal variation can be described as

$$\text{temporal delay difference} = P_2 - S_2. \quad (2)$$

The differential delay from the time-step method in Equation (1) can be divided into two parts; temporal delay difference and spatial differential delay.

$$dI_{time-step method} = P_2 - S_1 = P_2 - S_2 + S_2 - S_1 \quad (3)$$

$P_2 - S_2$: temporal delay difference

$S_2 - S_1$: spatial delay difference

Therefore, if we get the differential ionospheric delay ($S_2 - S_1$) using the station-pair method, we can obtain the temporal delay difference ($P_2 - S_2$) as follows:

$$P_2 - S_2 = (P_2 - S_1)_{dI \text{ with time-step method}} - (S_2 - S_1)_{dI \text{ with station-pair method}} \quad (4)$$

The procedure for calculating the temporal delay differences is shown in Figure 13. First, we search for station pairs which have baseline distances below 50 km. The station pairs whose IPP separation vector is almost the same with that of the time-step method in both magnitude (D) and direction (θ) are then selected. Finally, the temporal delay difference and temporal gradient are calculated for the selected pairs.

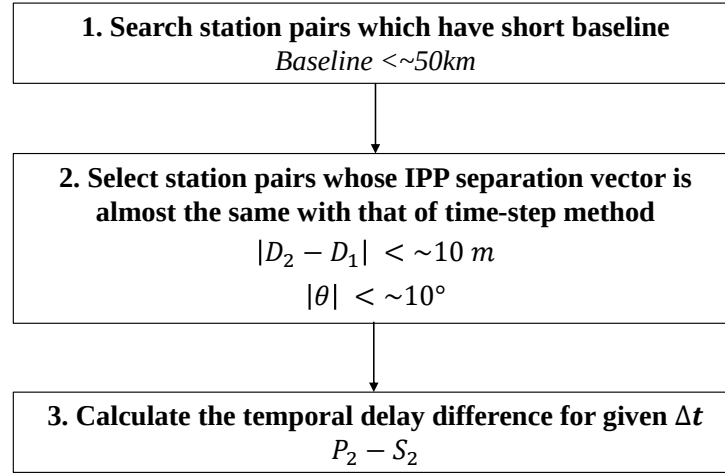


Figure 13: Procedure of the “geometric modeling”

However, the sparsity of the Brazilian GNSS observation networks makes it difficult to use this model to quantify temporal effects in the time-step method. Thus, we utilized “GNSS Earth Observation Network System (GEONET)” data from Japan, which have a dense GNSS observation network with an average baseline of about 20 km.

6.2. Results of Temporal Gradient Statistics

Plasma bubbles in equatorial region often cause large ionospheric gradients and these events are usually associated with ionospheric scintillation. To obtain statistical information of ionospheric temporal gradients under ionospheric scintillation conditions, which is the case for daytime in the Brazilian region, we chose a scintillating day, 1 October, 2010 (2010-274), on which several anomalous gradients as large as 200 mm/km were observed at Ishigaki Island in Japan [9] and analyzed GEONET data on this date.

The left plot of Figure 14 shows the temporal delay difference (in meters) as a function of IPP separation distance. Different colors for each temporal delay difference indicate different sizes of time window used for the time-step method. The right plot of Figure 14 shows the temporal gradient (in meters per minutes) which was computed by dividing the temporal delay difference by the corresponding time window to estimate ionospheric temporal decorrelation in a time interval of one minute. The temporal delay difference increases as the IPP separation distance increases, because its corresponding time interval between the ionosphere states at t_1 and at t_2 gets longer. When the temporal delay difference was divided by its time window value, the relatively flat tendency of the temporal gradient regardless of IPP separation distance was observed.

Table 5 shows the statistics of temporal gradients for a time interval of one-minute on 1 October, 2010. The mean and standard deviation of temporal gradient estimates are 0.0001 m/min and 0.0077 m/min respectively. When we consider IPP separation distances of about 4 to 6 km (a typical range which corresponds to one-minute time window), the standard deviation of 0.0077 m/min can be converted to $1.3\sim 1.9 \text{ mm/km} = 7.7 \text{ mm} / 4\sim 6 \text{ km}$. Therefore, the one-sigma of temporal gradient estimates is about 2 mm/km, when a time window of one minute is used for the time-step method. While it is not practical to remove the temporal gradient effect from the overbounding σ_{vig} derived in Section 4, the results from estimation of temporal gradients using the proposed method indicates that the σ_{vig} estimates derived using the time step method include the component of temporal gradients and thus somewhat conservative.

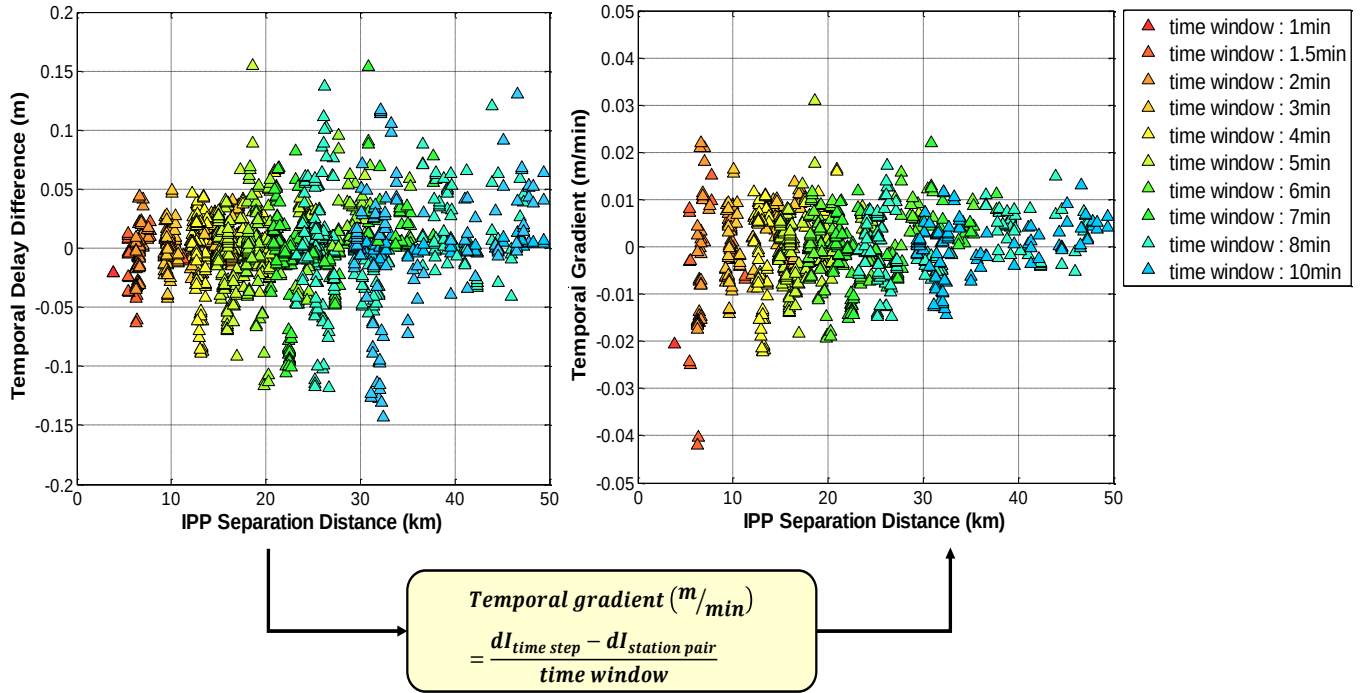


Figure 14: Temporal delay difference (left) and temporal gradient (right) vs. IPP separation distance

Table 5. Statistics of temporal gradient on 1 October, 2010

Number of Data	Mean (m/min)	Standard deviation (m/min)
1037	0.0001	0.0077

7.0 SUMMARY AND REMAINING STEPS

This paper has demonstrated that, while the value of σ_{vig} that bounds nominal ionospheric spatial gradients has been determined to be approximately 4 mm/km in mid-latitude CAT I GBAS operations, a significantly higher value should be broadcast for σ_{vig} at GIG to bound low-latitude ionospheric conditions. The results in Section 4 demonstrate that a minimum broadcast σ_{vig} value of 14 mm/km (including both ionospheric and tropospheric components) is sufficient to bound nominal ionospheric behavior during daytime, where “daytime” is shown to generally apply to the period between 6 AM and 6 PM Local Time, or, more particularly, between local sunrise and local sunset. This paper also evaluates the effect of ionospheric temporal gradients included in the estimates of ionospheric spatial gradient computed using the time-step method. With the proposed “geometric modeling” method, the one-sigma of temporal gradients for one-minute interval was estimated to be about 2 mm/km.

Because the times of local sunrise and sunset vary both in terms of local time and GPS Time (which is observed by the SLS-4000), a practical means for the initial implementation of SLS-4000 operations at Rio to determine when to switch to and from “daytime” conditions (and thus to activate or deactivate GBAS approach service) is needed. The beginnings of an approach to do this is proposed in Section 4.0 but needs review and further consideration. A key issue is how often these switching times (in terms of GPS Time) can change over the course of one year to compensate for variations in local sunrise and sunset.

The results should be sufficient to approve and certify CAT I precision approach operations using the existing SLS-4000 in Rio during the daytime hours defined here. In future work, the same analysis tools can be used to determine how to extend GBAS operations into the early morning and early evening periods while maintaining integrity against the more-severe ionospheric conditions that exist outside daytime hours. A re-examination of the seasonality of the gradients that motivate the proposed σ_{vig} bound is recommended, as it may be possible to lower this bound and/or extend the definition of daytime hours during the middle of the year, when large gradients are generally not observed.

ACKNOWLEDGMENTS

The authors gratefully acknowledge Shelly Beauchamp of the FAA William J. Hughes Technical Center and her team for their support. We would like to thank John Warburton, Jason Burns, and Randy Key of the FAA and Oliver Jeannot, Dieter Guenter, and Leo Eldredge of Tetra Tech AMT for their support of this work. The help and support of many people working with the Brazilian Department of Airspace Control (DECEA) to facilitate GBAS operations in Brazil is also greatly appreciated, including Rich Cole and Navin Mathur of MIRUS Corp., Pat Doherty, Rezy Pradipta, and Endawoke Yizengaw of Boston College, Brian Koosman of Honeywell, Eurico Paula and Jonas Souza of the Brazilian National Institute for Space Research (INPE), and Alessandro Santoro of DECEA. Hyeyeon Chang was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (NRF-2018R1D1A1A09082517). Moonseok Yoon was supported by the Space Core Technology Development Program of the National Research Foundation (NRF) funded by the Ministry of Science, ICT & Future Planning (NRF-2014M1A3A3A02034937).

REFERENCES

1. M. Yoon, J. Lee, S. Pullen, *et al*, “Equatorial Plasma Bubble Threat Parameterization to Support GBAS Operations in the Brazilian Region,” *Navigation*, Vol. 64, No. 3, Fall 2017, pp. 309-321.
2. J. Lee, and S. Pullen, “Anomalous Ionospheric Gradients over Brazil during Local Daytime,” Brazil GBAS Safety Case, DECEA Meeting, Rio de Janeiro, Brazil, 19 March 2018.
3. J. Lee, S. Pullen, S. Datta-Barua and P. Enge (2007), “Assessment of Ionosphere Spatial Decorrelation for Global Positioning System-Based Aircraft Landing Systems,” *Journal of Aircraft*, vol. 44, no. 5, pp. 1662-1669.
4. Huang, C.-S., O. de La Beaujardiere, P. A. Roddy, D. E. Hunton, J. O. Ballenthin, and M. R. Hairston (2013), “Long-lasting daytime equatorial plasma bubbles observed by C/NOFS satellite”, *Journal of Geophysical Research*, vol. 118, pp. 2398-2408
5. S. Datta-Barua, J. Lee, *et al*, “Ionospheric Threat Parameterization for Local Area Global-Positioning-System-Based Aircraft Landing Systems,” *AIAA J. of Aircraft*, Vol. 47, No. 4, July-Aug. 2010, pp. 1141-1151.
6. Moonseok Yoon, Dongwoo Kim, *et al*, “Multi-dimensional Verification Methodology of Ionospheric Gradient Observation during Plasma Bubble Events in the Brazilian Region,” *Proceedings of ION Pacific PNT 2015*, Honolulu, HI, April 20-23, 2015, pp. 748-762.
7. “Sunrise and Sunset Times in Rio de Janeiro,” timeanddate.com, <http://www.timeanddate.com/sun/brazil/rio-de-janeiro>.
8. “Sun or Moon Rise/Set Table for One Year,” U.S. Naval Observatory (USNO) Astronomical Applications Dept., http://aa.usno.navy.mil/data/docs/RS_OneYear.php.
9. S. Saito, S. Fujita, and T. Yoshihara, “Precise measurements of ionospheric delay gradient at short baselines associated with low latitude ionospheric disturbances,” *Proceedings of the 2012 International Technical Meeting (ITM) of The Institute of Navigation*, Newport Beach, CA, January 2012, pp. 1445-1450.